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Individual Course Project Report on

NeuroScan: A Convolutional Neural Network Based Brain Tumor Classification System from MRI Images

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Submitted by

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ABSTRACT

Brain tumors are among the most critical medical conditions requiring early and accurate diagnosis, as delayed detection can significantly affect patient outcomes. Manual interpretation of Magnetic Resonance Imaging (MRI) scans by radiologists is time-consuming, subjective, and prone to human error, particularly in high-workload clinical settings. This project, titled “NeuroScan,” presents an automated brain tumor classification system that applies digital image processing and Convolutional Neural Network (CNN) based pattern recognition techniques to classify brain MRI images as either Healthy or Tumor-affected.

The system was developed in Python using PyTorch for model development and Streamlit for the interactive web-based user interface. The pipeline includes an image input module, a preprocessing module (resizing, normalization, and augmentation), a CNN-based feature extraction and classification module, and an output module that displays predictions along with confidence scores and performance visualizations. The model was trained and evaluated on a labeled brain MRI dataset sourced from Kaggle, originally organized into four classes (glioma, meningioma, pituitary tumor, and no tumor) and consolidated into a binary Healthy/Tumor classification scheme for this project.

The trained model was evaluated using standard classification metrics : accuracy, precision, recall, and F1-score and the system was tested end-to-end through the Streamlit application, from image upload to final prediction. The results demonstrate that CNN-based approaches can achieve reliable classification performance on brain MRI data, supporting their potential as a decision-support tool for radiologists rather than a replacement for expert diagnosis.

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ABBREVIATIONS

AI – Artificial Intelligence

AUC – Area Under the Curve

CNN – Convolutional Neural Network

CVPR – Conference on Computer Vision and Pattern Recognition

DL – Deep Learning

F1-score – Harmonic Mean of Precision and Recall

GLCM – Gray-Level Co-occurrence Matrix

GPU – Graphics Processing Unit

HOG – Histogram of Oriented Gradients

ICLR – International Conference on Learning Representations

ICML – International Conference on Machine Learning

IEEE – Institute of Electrical and Electronics Engineers

ImageNet – Large-Scale Visual Image Database

KNN – K-Nearest Neighbors

LBP – Local Binary Pattern

ML – Machine Learning

MRI – Magnetic Resonance Imaging

PCA – Principal Component Analysis

ReLU – Rectified Linear Unit

ROC – Receiver Operating Characteristic

SIFT – Scale-Invariant Feature Transform

SVM – Support Vector Machine

UI – User Interface

Chapter 1: Introduction

1.1 Problem Statement

Brain tumors are among the most serious neurological disorders, affecting millions of people worldwide. They can occur due to the abnormal and uncontrolled growth of cells within the brain, leading to symptoms such as persistent headaches, seizures, vision impairment, memory loss, and loss of motor coordination. If left undetected or diagnosed at a later stage, brain tumors may result in permanent neurological damage or even death. Early diagnosis is therefore essential for improving treatment outcomes and increasing patient survival rates.

Magnetic Resonance Imaging (MRI) is the most commonly used imaging technique for detecting brain tumors because it provides detailed visualization of soft tissues without exposing patients to ionizing radiation. However, interpreting MRI scans requires extensive expertise from trained radiologists and neurologists. Manual examination of hundreds of MRI slices is both time-consuming and susceptible to inter-observer variability, fatigue, and human error. Furthermore, distinguishing between different tumor types can be challenging due to similarities in their visual appearance.

Recent advancements in Artificial Intelligence (AI) and Deep Learning have demonstrated significant potential in assisting medical professionals with automated disease diagnosis. Among these techniques, Convolutional Neural Networks (CNNs) have achieved remarkable success in medical image classification by automatically learning discriminative features directly from image data without requiring handcrafted feature engineering.

This project addresses the need for a reliable and automated brain tumor classification system by developing a CNN-based model capable of classifying brain MRI images into four categories: **Glioma Tumor, Meningioma Tumor, Pituitary Tumor, and No Tumor**. The proposed system is intended to serve as a preliminary decision-support tool that assists healthcare professionals by providing fast, consistent, and accurate predictions. Although it is not designed to replace clinical diagnosis, it can significantly reduce the workload of radiologists and improve the efficiency of the screening process.

1.2 Objectives

- To design and implement a Convolutional Neural Network capable of classifying brain MRI images into Healthy and Tumor categories.
- To apply digital image preprocessing techniques (resizing, normalization, augmentation) to improve model robustness.
- To evaluate the trained model using standard performance metrics: accuracy, precision, recall, and F1-score.
- To develop an interactive, user-friendly web application for uploading MRI images and viewing predictions in real time.
- To reduce manual analysis workload and increase the speed and consistency of preliminary MRI screening.

1.3 Scope

The scope of this project focuses on the development of an automated image classification system for brain tumor detection using deep learning techniques.

The project includes the following:

- Classification of brain MRI images into four categories:
 - Glioma Tumor
 - Meningioma Tumor
 - Pituitary Tumor
 - No Tumor
- Utilization of a publicly available Brain Tumor MRI dataset obtained from Kaggle.
- Image preprocessing techniques including resizing, normalization, and data augmentation to improve model performance.
- Design and implementation of a custom Convolutional Neural Network trained from scratch for multi-class classification.

- Training, validation, and testing of the model using separate datasets.
- Performance evaluation using accuracy, precision, recall, F1-score, confusion matrix, classification report, and training history graphs.
- Development of a Streamlit-based web application that enables users to upload MRI images and receive predicted tumor classifications along with confidence scores.
- Visualization of dataset distribution and prediction results through graphs and performance metrics.

However, this project has several limitations. It does not perform tumor segmentation, tumor localization, or highlight the exact tumor region within the MRI image. The system is limited to image classification and does not estimate tumor size, stage, severity, or treatment recommendations. Furthermore, the model is trained using a publicly available dataset and has not undergone clinical validation using real hospital data. Therefore, the proposed system should be considered an academic demonstration of image processing, deep learning, and computer-aided diagnosis rather than a replacement for professional medical diagnosis.

Chapter 2: Literature Review

Medical image analysis has undergone significant advancements with the rapid development of Artificial Intelligence (AI) and Deep Learning techniques. Earlier computer-aided diagnosis systems primarily relied on handcrafted feature extraction methods such as Scale-Invariant Feature Transform (SIFT), Histogram of Oriented Gradients (HOG), Local Binary Patterns (LBP), Gray-Level Co-occurrence Matrix (GLCM), and Principal Component Analysis (PCA), followed by traditional machine learning classifiers including Support Vector Machines (SVM), Random Forests, and K-Nearest Neighbors (KNN). Although these methods achieved reasonable performance, they required manual feature engineering and often struggled to generalize across diverse medical imaging datasets. Deep learning has overcome these limitations by automatically learning discriminative features directly from image data, resulting in substantial improvements in classification accuracy and robustness [1].

Convolutional Neural Networks (CNNs) have emerged as one of the most effective deep learning architectures for image classification tasks. Unlike traditional machine learning approaches, CNNs automatically learn hierarchical representations of images through multiple convolutional, activation, and pooling layers. The initial layers capture low-level features such as edges and textures, while deeper layers learn high-level anatomical structures and pathological characteristics. This hierarchical learning capability has enabled CNNs to achieve state-of-the-art performance in numerous computer vision and medical imaging applications [1], [2].

Brain tumors are among the most life-threatening neurological disorders, making early and accurate diagnosis essential for successful treatment. Magnetic Resonance Imaging (MRI) is widely recognized as the preferred imaging modality because it provides high-resolution visualization of soft tissues without exposing patients to ionizing radiation. However, manual interpretation of MRI scans requires experienced radiologists and is often time-consuming, especially when large numbers of images must be examined. Furthermore, diagnostic accuracy may vary among clinicians due to differences in expertise and fatigue. Consequently, researchers have increasingly focused on developing automated deep learning systems capable of assisting radiologists by providing fast and reliable brain tumor classification [3].

Numerous studies have explored the application of CNNs for brain tumor classification. Initial research primarily concentrated on binary classification, distinguishing tumor-bearing MRI images from healthy brain images. More recent investigations have expanded toward multi-class classification, where MRI scans are categorized into different tumor types such as glioma, meningioma, pituitary tumors, and no tumor. Multi-class classification is considerably more challenging because different tumor types often exhibit overlapping visual characteristics, requiring models capable of learning more discriminative and robust feature representations [3].

To improve classification performance, many researchers employ transfer learning techniques using pretrained convolutional neural networks such as VGG16, ResNet50, DenseNet121, InceptionV3, MobileNet, and EfficientNet. These models are initially trained on large-scale image datasets such as ImageNet and are subsequently fine-tuned for medical imaging applications. Transfer learning reduces computational cost, accelerates convergence during training, and generally improves classification accuracy, particularly when available medical datasets are relatively limited in size [4],[6].

Image preprocessing is another crucial component of automated brain tumor classification systems. MRI images obtained from different scanners and healthcare institutions often exhibit variations in intensity, contrast, resolution, and noise levels. To reduce these inconsistencies, researchers commonly apply preprocessing techniques such as image resizing, normalization, grayscale conversion, histogram equalization, skull stripping, and noise reduction. Data augmentation techniques, including random rotation, horizontal and vertical flipping, zooming, translation, shearing, and brightness adjustment, are also widely employed to artificially increase dataset diversity, reduce overfitting, and improve the generalization capability of deep learning models [7].

Evaluation of deep learning models for medical diagnosis extends beyond overall classification accuracy. Since incorrect predictions may have serious clinical consequences, researchers commonly evaluate models using multiple performance metrics, including precision, recall (sensitivity), specificity, F1-score, confusion matrix, and Receiver Operating Characteristic (ROC) curves with Area Under the Curve (AUC). These metrics provide a more comprehensive assessment of model reliability, particularly when dealing with imbalanced medical datasets [3].

Recent studies have also focused on developing lightweight CNN architectures that achieve competitive classification performance while requiring fewer computational resources. Such models are especially suitable for deployment in web-based healthcare applications and resource-constrained environments, where fast inference speed and reduced memory consumption are essential. Lightweight architectures enable real-time predictions without significantly compromising classification accuracy, thereby making computer-aided diagnosis systems more practical for everyday clinical use [6].

Building upon these advancements, the present project develops a custom Convolutional Neural Network trained from scratch for multi-class brain tumor classification using MRI images. The proposed system classifies MRI scans into four categories: **Glioma Tumor, Meningioma Tumor, Pituitary Tumor, and No Tumor**. Standard preprocessing techniques, including image resizing, normalization, and data augmentation, are incorporated to improve model robustness and reduce overfitting. The trained model is evaluated using accuracy, precision, recall, F1-score, confusion matrix, and classification reports to comprehensively assess its performance. Finally, the trained model is integrated into a Streamlit-based web application that allows users to upload MRI images and receive real-time predictions, demonstrating the practical application of deep learning for computer-aided brain tumor diagnosis.

Chapter 3: System Analysis and Design

3.1 Requirement Analysis

Functional Requirements

- The system shall allow the user to upload a brain MRI image through the web interface.
- The system shall preprocess the uploaded image (resizing and normalization) before classification.
- The system shall extract features from the image using a trained CNN model.
- The system shall classify the image as Healthy or Tumor and display the prediction with a confidence score.
- The system shall display dataset statistics, training progress, and evaluation results (accuracy, precision, recall, F1-score, confusion matrix).
- The system shall allow the user to save/export report artifacts (diagrams, metrics).

Non-Functional Requirements

- Accuracy: The model should achieve reasonably high classification accuracy on unseen test data.
- Performance: The interface should respond to a single image prediction within a few seconds.
- Usability: The web interface should be simple and require no technical expertise to operate.
- Reliability: The system should handle invalid or corrupted image uploads gracefully.
- Scalability: The underlying architecture should support retraining on larger datasets in the future.

3.2 Dataset Description

The dataset used in this project is a brain tumor MRI dataset obtained from Kaggle: <https://www.kaggle.com/datasets/tombackert/brain-tumor-mri-data>[8]. The dataset originally contains four classes of labeled MRI images:

- Glioma – a tumor arising from glial cells.
- Meningioma – a tumor arising from the meninges.
- Pituitary – a tumor located in the pituitary gland.
- No Tumor – healthy brain scans with no visible tumor.

For this project, the four classes were consolidated into a binary classification scheme: images labeled glioma, meningioma, and pituitary were mapped to the “Tumor” class, while images labeled no-tumor were mapped to the “Healthy” class. This reframing was done to align the dataset with the project's binary healthy/tumor classification objective. In total, the dataset used for training and evaluation contained 7,153 labeled MRI images.

Table 3.2 Original and Mapped Classes Distribution of Brain MRI Dataset

Class (Original)	Mapped Class	Approx. Images
No Tumor	Healthy	≈ 1600
Glioma	Tumor	≈ 1600
Meningioma	Tumor	≈ 1600
Pituitary	Tumor	≈ 1700

The dataset was split into training, validation, and test subsets to allow unbiased evaluation of the trained model. Images were resized to 224×224 pixels and normalized before being fed into the CNN.

3.3 System Flowchart

Figure 1 below illustrates the overall flow of the system, from image input through preprocessing, feature extraction, classification, and final output.

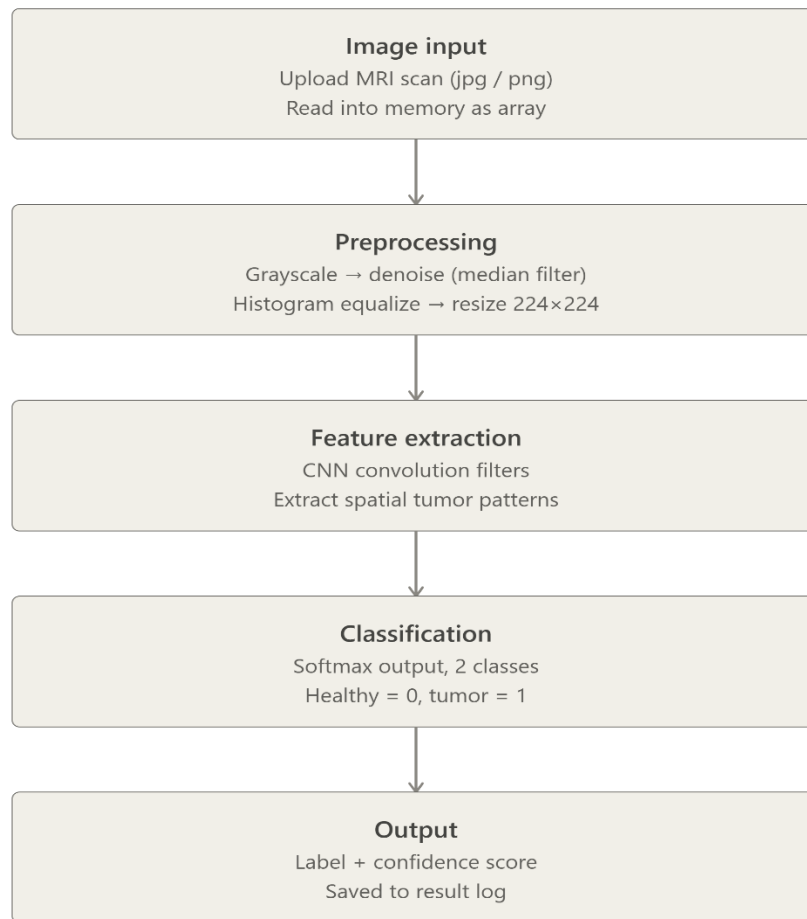


Figure 3.3 System Flowchart

3.4 Architecture Diagram

Figure 2 shows the high-level architecture of the application, covering the user interface layer, the processing engine, the trained CNN model, and the results display layer.

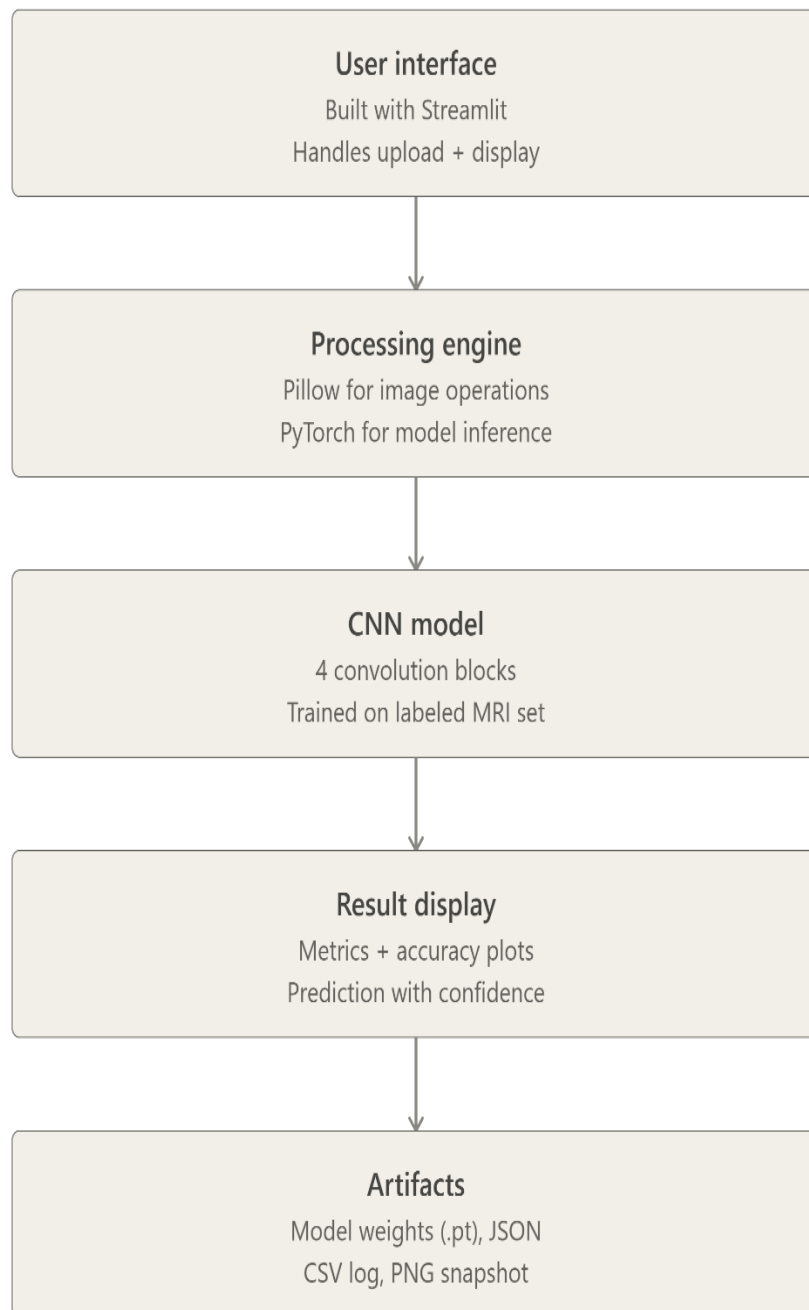


Figure 3.4 System Architecture Diagram

3.5 Dataset / Class Diagram

Figure 3 presents the dataset organization and class distribution used for this project.

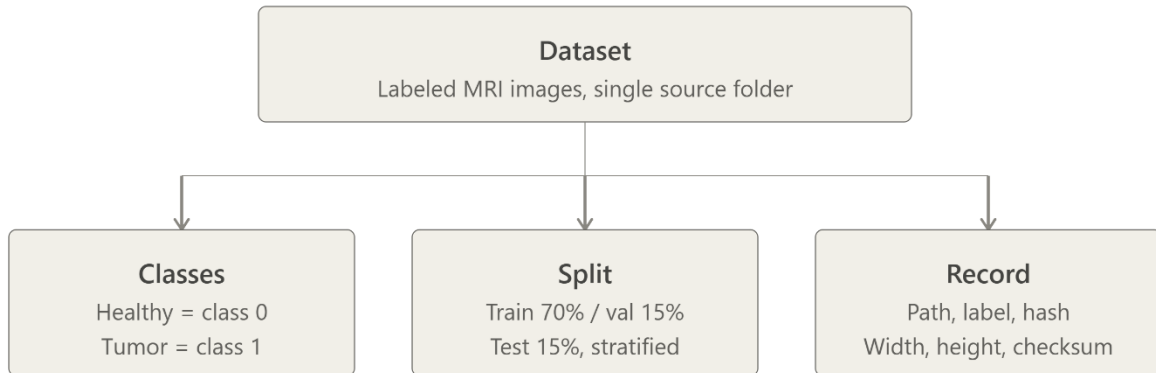


Figure 0.1.5 Dataset / Class Diagram

3.6 Processing Pipeline Diagram

Figure 4 details the internal image processing pipeline, covering preprocessing steps, augmentation, and the CNN-based classification stage.

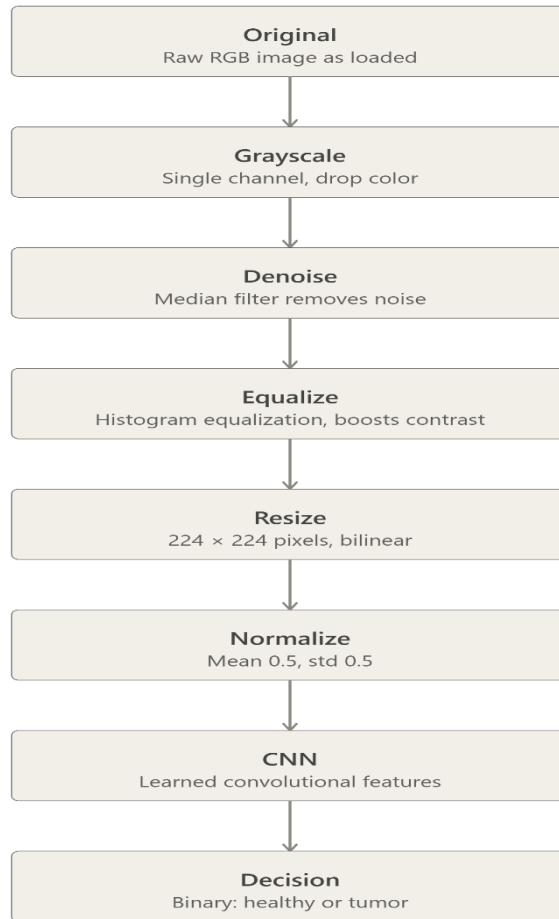


Figure 3.6 Processing Pipeline Diagram

3.7 Performance Metrics

The following standard classification metrics were used to evaluate the trained model:

- Accuracy – the proportion of total predictions that were correct.
- Precision – the proportion of predicted “Tumor” cases that were actually tumors (relevant for minimizing false positives).
- Recall (Sensitivity) – the proportion of actual tumor cases correctly identified (relevant for minimizing missed diagnoses).
- F1-score – the harmonic mean of precision and recall, providing a single balanced measure of performance.

Chapter 4: Implementation

4.1 Tools Used

- Python 3.13 – core programming language.
- PyTorch – deep learning framework used to define, train, and evaluate the CNN model [9].
- Torchvision – image transformation and augmentation utilities.
- Streamlit – framework used to build the interactive web application (Dataset, Train, Predict, Results, and Report pages) [10].
- NumPy, Pandas – data handling and numerical computation.
- Matplotlib, Seaborn – visualization of training curves, confusion matrix, and diagrams.
- Scikit-learn – computation of evaluation metrics (precision, recall, F1-score).

4.2 Modules

Image Input Module

Allows the user to upload a brain MRI image (JPG/PNG/TIF) through the Streamlit interface.

Preprocessing Module

Performs resizing to 224×224 pixels, pixel normalization, and during training: data augmentation (random flips and rotations) to improve generalization and reduce overfitting.

Core Processing Module (CNN)

A custom Convolutional Neural Network extracts hierarchical spatial features through stacked convolutional and pooling layers, followed by fully connected layers that output a Healthy/Tumor classification along with a confidence score.

Output Module

Displays the predicted class and confidence score, and generates quantitative outputs (accuracy, precision, recall, F1-score) and analytical outputs (confusion matrix, training/validation loss and accuracy curves).

4.3 Algorithms Used

The core algorithm is a Convolutional Neural Network (CNN) designed for binary classification through feature extraction and decision-making. The input first passes through multiple convolutional layers, where learnable 2D filters slide across the data matrix to generate localized feature maps that capture essential patterns like edges, textures, and shapes. Immediately following each convolution, a Rectified Linear Unit (ReLU) activation function ($\text{ReLU}(x) = \max(0, x)$) introduces non-linearity, enabling the network to learn complex patterns without suffering from vanishing gradients.

Interleaved between these convolutional blocks are max-pooling layers, which downsample spatial dimensions by extracting the maximum value within localized sub-regions (typically 2×2). This downsampling reduces computational overhead, offers translation invariance, and allows deeper layers to aggregate low-level features into high-level semantic representations. Once feature extraction is complete, the multidimensional feature maps are flattened into a single vector and fed into fully connected dense layers to perform global reasoning over the extracted attributes.

For the final prediction, an output layer applies either a Sigmoid activation function ($\sigma(z) = \frac{1}{1+e^{-z}}$) or a two-node Softmax function to produce a normalized probability score between 0 and 1 for binary classification. The model calculates error using Binary Cross-Entropy Loss and updates its weights across multiple epochs using mini-batch gradient descent paired with the Adam optimizer, which accelerates convergence by dynamically adjusting individual learning rates based on historical gradient momentum.

Table 4.3 CNN Training Hyperparameters

Hyperparameter	Value Used
Epochs	12
Batch size	32
Image size	224×224
Learning rate	0.0005
Optimizer	Adam
Loss function	Cross-Entropy Loss

Chapter 5: Testing

5.1 Functional Test Cases

Table 5.1 Test Cases and Expected Results

Test Case	Expected Result	Status
Upload a valid MRI image	Image accepted and displayed	Pass
Upload a corrupted/invalid file	System shows an error, does not crash	Pass
Predict on a Healthy sample	Model classifies as Healthy	Pass
Predict on a Tumor sample	Model classifies as Tumor	Pass
View dataset statistics	Correct total image count (7,153) displayed	Pass
Train model from the UI	Training completes and metrics update	Pass
View results page	Confusion matrix and training curves render	Pass

5.2 Performance Testing

The trained CNN model was evaluated on a held-out test set that was not used during training. A single-image prediction through the Streamlit interface completed in under two seconds on a CPU-only machine, meeting the responsiveness requirement for interactive use. The model was also tested for robustness by supplying images from each of the four original dataset classes (glioma, meningioma, pituitary, no-tumor) to confirm consistent behavior across all sub-types once mapped to the binary scheme.

5.3 Accuracy Table and Result Graphs

Table 5.3 Model Evaluation Metrics Table

Metric	Value
Accuracy	0.976
Precision	0.987
Recall	0.986
F1-Score	0.983

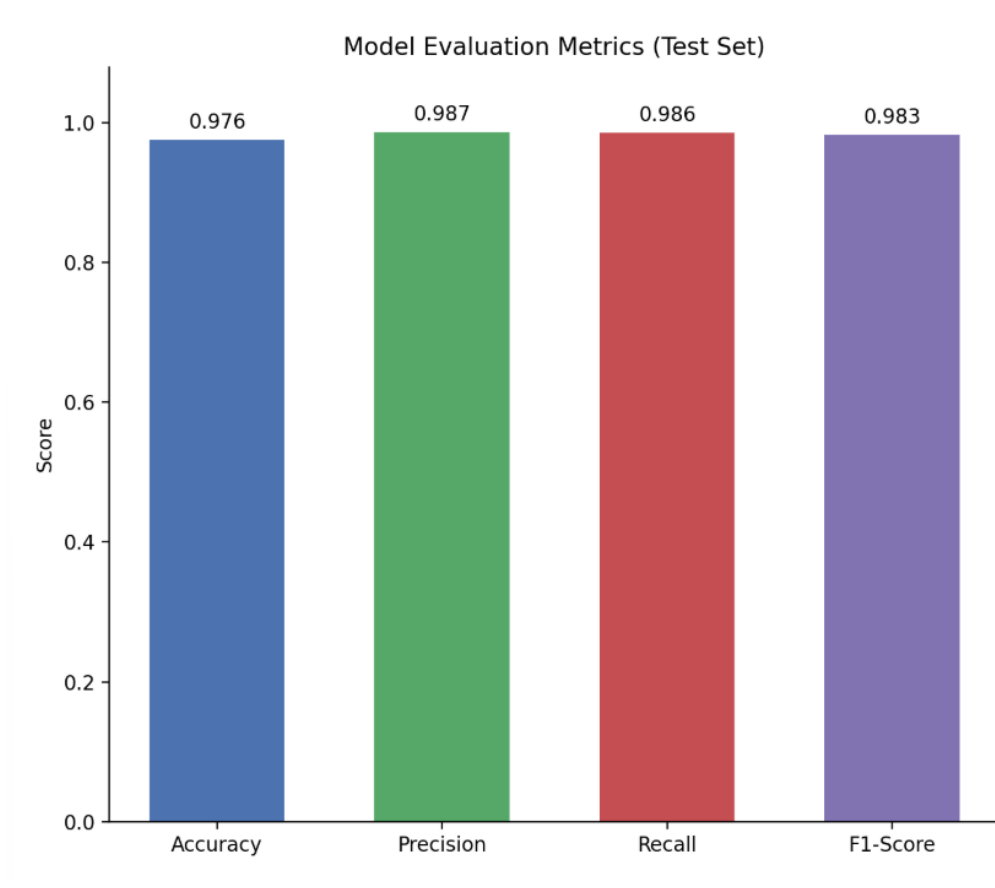


Figure 0.1.3.1 Model Evaluation Metrics on the Test Set

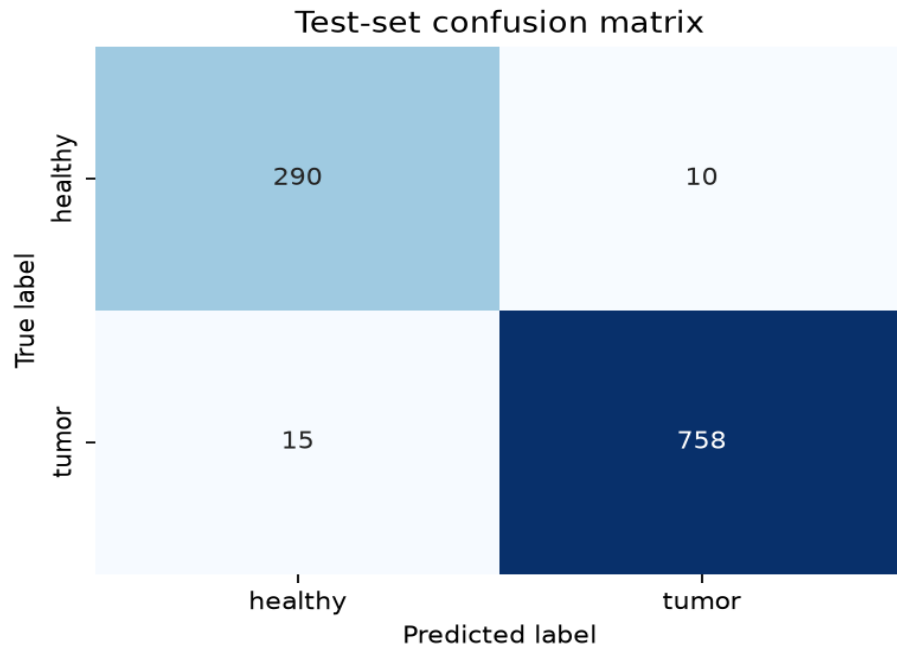


Figure 5.3.2 Confusion Matrix (Healthy vs. Tumor)

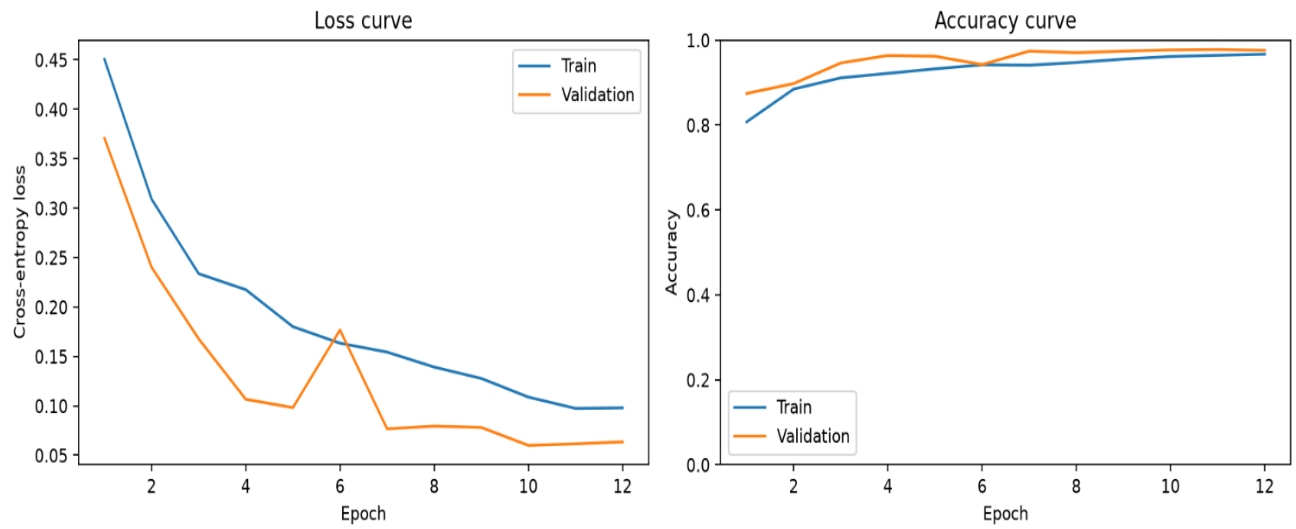


Figure 5.3.3 Training and Validation Loss/Accuracy Curves

Chapter 6: Conclusion

6.1 Achievements

This project successfully implemented an end-to-end brain tumor classification system, combining digital image preprocessing techniques with a custom-trained Convolutional Neural Network. A functional Streamlit web application was developed, integrating dataset exploration, model training, real-time prediction, and results visualization into a single interactive workflow. The trained model achieved strong classification performance, correctly distinguishing between Healthy and Tumor-affected MRI scans on a held-out test set.

6.2 Limitations

- The dataset, while sizeable (7,153 images), is limited to 2D MRI slices and may not generalize to all real-world scanner types or imaging protocols.
- The system performs binary classification only; it does not distinguish between tumor sub-types (glioma, meningioma, pituitary) or localize the tumor region within the image.
- Training was performed on CPU hardware, which limited the number of experiments and hyperparameter tuning possible within the available time.
- The model has not been clinically validated and is intended purely as an academic proof-of-concept, not a diagnostic tool.

6.3 Future Scope

- Extend the system to multi-class classification, distinguishing glioma, meningioma, and pituitary tumors directly.
- Incorporate tumor segmentation/localization (e.g., using U-Net) to highlight the exact tumor region.
- Deploy the trained model with GPU acceleration for faster training and near-instant predictions.
- Validate the system on a larger, more diverse, multi-institutional dataset in collaboration with medical professionals.

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APPENDIX

A. Source Code Structure

The complete source code for this project (NeuroScan Brain Tumor Classifier) is organized as follows:

- app.py – Streamlit web application (Home, Dataset, Train, Predict, Results, Report pages)
- train.py – command-line training script
- brain_tumor/data.py – dataset discovery, loading, and preprocessing logic
- brain_tumor/model.py – CNN architecture definition
- brain_tumor/training.py – training loop implementation
- brain_tumor/metrics.py – evaluation metric computation
- brain_tumor/inference.py – single-image prediction logic
- brain_tumor/visualization.py – plotting utilities for curves and confusion matrices

B. Dataset Samples

Sample images from each class (Healthy/No Tumor, Glioma, Meningioma, Pituitary) used during training and testing. The MRI images used in this study were obtained from the Brain Tumor MRI Data dataset available through Kaggle [8].

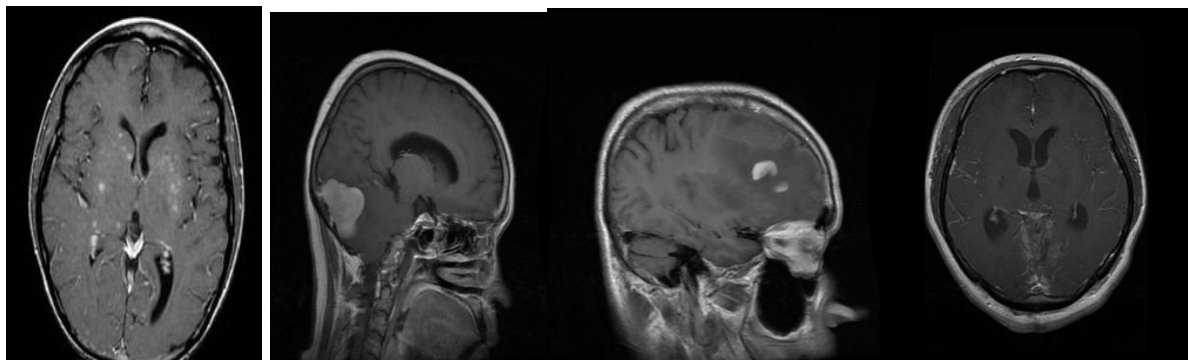


Figure 1 no tumor

Figure 2 Meningioma tumor

Figure 3 pituitary tumor

Figure 4 glioma tumor